MIL-W-2129A
15 August 1970
SUPERSEDING
MIL-S-2129
31 March 1950

MILITARY SPECIFICATION

WHEEL SET, ABRASIVE, ALUMINUM OXIDE,
MANDREL-MOUNTED, DENTAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a set of mandrel-mounted aluminum oxide abrasive wheels suitable for dental use.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply:

U.S. DEPARTMENT OF COMMERCE
NATIONAL BUREAU OF STANDARDS
COMMERCIAL STANDARD

CS 271-65	Grading of Abrasive Grain for Grinding Wheels

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive. The abrasive shall be 240-grit aluminum oxide conforming to the applicable requirements of CS 271-65 for grading of grain. The aluminum oxide shall be clean and shall be free from all extraneous materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

3.1.2 Bonding agent. The agent for bonding the aluminum oxide abrasive to the mandrel head shall be high quality, vitrified pipe clay (ball clay) ceramic.

3.1.3 Mandrel. The mandrel shall be fabricated from AISI No. 416 corrosion-resisting steel conforming to FED-STD-66 for chemical composition.

3.1.4 Cement. The cement shall be a phenolic or an epoxy resin cement suitable for permanently mounting the abrasive wheel head onto the mandrel.

3.2 Style, design, and dimensions. The set of mandrel-mounted, aluminum oxide abrasive wheels shall contain six different-sized, fine-grit hones suitable for use during the honing and sharpening of high-carbon steel and corrosion-resisting steel dental cutting instruments. Each abrasive wheel set shall consist of one each of sizes 5, 7, 9, 11, 14 and 17 wheels. The head and mandrel dimensions of each of the six sizes required in the set shall be as specified in Table I. The head of each size of wheel shall be cylindrical and untapered (square-edged). The length of each mandrel prior to assembly shall be 1-5/8 ± 1/16 inches.

Table I. Dimensions of wheels in set

Size	Length of head (± 0.015 inch)	Diameter of head (± 0.015 inch)	Diameter of mandrel (± 0.0005 inch)	Length of exposed mandrel (± 1/16 inch)
5	0.325	0.205	0.0924	1-1/2
7	0.285	0.217	0.0924	1-1/2
9	0.310	0.366	0.0924	1-1/2
11	0.283	0.457	0.0924	1-1/2
14	0.285	0.543	0.0924	1-1/2
17	0.315	0.675	0.0924	1-1/2

3.3 Construction. The aluminum oxide abrasive particles and vitrified pipe clay shall be uniformly mixed and properly heat-treated so as to result in a finished head having a homogeneous appearance. The proximal end of the mandrel shall be rounded, and the distal end shall be properly knurled or serrated to receive the abrasive head which is to be mounted thereon. The head shall be securely and permanently attached to the concealed end of the mandrel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

3.4 Finish. The mandrels on the abrasive wheels shall have a bright polished finish.

3.5 Corrosion resistance. The mandrels on the abrasive wheels shall pass the boil test and copper sulfate test for corrosion resistance, when tested as specified in 4.4.2.

3.5.1 Resistance to boil. All exposed surfaces of the mandrels shall be smooth and shall not show any signs of corrosion.

3.5.2 Resistance to copper sulfate. There shall be no plating of copper on the mandrels. Any dulling of the polished surfaces, however, shall be disregarded.

3.6 Runout. The total indicated runout of the abrasive wheels shall be no greater than 0.004 inch, when determined as specified in 4.3.1.

3.7 Performance. The surface roughness on the honed and sharpened edges of dental cutting instruments shall be no greater than 30 micro inches, when the abrasive wheels are tested as specified in 4.4.3.

3.8 Workmanship. Workmanship shall be first class throughout. The aluminum oxide abrasive wheel sets shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables II and IV (see 4.2.1 and 4.3) and Table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table II. The unit of product for examination purposes shall be one complete set of six abrasive wheels.

Table II. Examination criteria

Type of defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III, except that the acceptance number shall be zero for all sample sizes. The unit of product for the corrosion resistance test shall be one finished wheel or one mandrel, prior to assembly; and for the performance test, shall be one complete set of finished wheels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

Table III. Test criteria

Component	Characteristic	Requirement	Test procedure	Inspection level
Abrasive particles	Grading of grain	3.1.1	CS 271-65	(See 4.2.3)
Raw steel	Chemical composition	3.1.3	FED-STD-66 and 4.4.1	(See 4.2.4)
Finished wheel, or mandrel prior to assembly	Corrosion resistance:	3.5	4.4.2	- - - -
	Boil test	3.5.1	4.4.2.1	S-1
	Copper sulfate test	3.5.2	4.4.2.2	S-1
Finished wheel	Performance	3.7	4.4.3	S-1

4.2.3 Sampling for abrasive particle grading. A sample of 100.0 ± 5.0 grams from each lot or batch of abrasive particles shall be selected for grading.

4.2.4 Sampling of raw materials. A sufficient quantity of each lot of raw material used in the manufacture of the abrasive wheel mandrels shall be selected for chemical analysis.

4.2.5 Sample unit usage. Where feasible, the same abrasive wheels shall be used for the determination of two or more test characteristics.

4.3 Examination. The complete abrasive wheel set shall be examined for defects including, but not limited to, those listed in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

Table IV. Classification of defects

Category	Defect
Major A:	
101	Style and design of any wheel not as specified.
102	Any incorrect size or sizes of abrasive wheels in set furnished; any size missing; more than one of same size furnished in any single set.
103	Head not securely and permanently attached to concealed end of mandrel.
Major B:	
151	Diameter of head not within range specified for applicable size.
152	Diameter of mandrel not 0.0924 ± 0.0005 inch.
153	Length of head not within range specified for applicable size.
154	Total indicated runout exceeds 0.004 inch.
Minor A:	
201	Finish on mandrel not as specified.
202	Proximal end of mandrel not rounded.
203	Mandrel not free from defects such as cracks, or pits.
204	Abrasive, as mounted on wheel, shows presence of extraneous material.
205	Packaging and packing not as specified.
206	Marking on packaging and pack incorrect, illegible, incomplete or missing altogether.
Minor B:	
251	Length of exposed portion of mandrel not as specified.
252	Cement covers an area exceeding 1/16 inch of mandrel above base of head.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

4.3.1 Runout. The mandrel of the abrasive wheel shall be placed and held in a "V" block or precision chuck, and rotated slowly by any suitable means. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the central axis of the abrasive wheel, and so positioned that the foot contacts the mandrel of the wheel at a point 1/8 ± 1/32 inch beyond the base of the head (where the mandrel contains no cement). The maximum and minimum readings obtained during one complete 360-degree rotation of the abrasive wheel shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.4 Test procedures.

4.4.1 Chemical composition. In the absence of certificates of quality for the raw steel used in the fabrication of the abrasive wheel mandrels, chemical analysis of the mandrels shall be determined in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.2 Corrosion resistance.

4.4.2.1 Boil test. Before testing, the exposed portion of the mandrels on the finished wheels, or the entire mandrels, prior to assembly, shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the mandrels shall be placed in a glass beaker containing distilled water, boiled for 30 ± 1 minutes, and then allowed to remain submerged for 24 ± 1/2 hours, during which time the water will cool to room temperature. At that time, the mandrels shall be removed from the beaker and examined for evidence of corrosion.

4.4.2.2 Copper sulfate test. Before testing, the exposed portion of the mandrels on the finished wheels, or the entire mandrels, prior to assembly, shall be scrubbed with soap and warm water; rinsed in hot water; dipped in either 95 percent ethyl alcohol, or National Formulary (NF) or American Chemical Society (ACS) isopropyl alcohol; and dried. A copper sulfate solution shall be made containing 4.0 ± 0.1 grams of copper sulfate pentahydrate, 10.0 ± 0.1 grams of sulfuric acid (sp. gr. 1.84), and 90.0 ± 1.0 milliliters of distilled water. The mandrels shall be immersed in the copper sulfate solution at a temperature of 65.0 ± 2.0 degrees Fahrenheit for 6.0 minutes ± 15 seconds. The mandrels shall then be wiped off using a cloth saturated with distilled water, and examined for presence of copper which may have become plated on the mandrel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

4.4.3 Performance. The abrasive wheel shall be used to hone and sharpen a dental cutting instrument fabricated from either AISI No. 440A, 440B, or 440C corrosion-resisting steel having a Rockwell hardness of C56 to C60. The surface roughness of the honed and sharpened instrument shall be determined by means of a Brush Surfindicator, or other equally suitable equipment.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. One wheel set, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be set into a perforated spacer of suitable material and then packaged in a box of appropriate size, constructed in accordance with PPP-B-676. Points shall be firmly held and so situated that they cannot contact each other or the walls of the container. Box shall be adequately closed to prevent accidental opening.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size, constructed in accordance with PPP-B-566, or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 8, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:26Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DH

Project No. 6520-1071

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-W-2129A FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE 47
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

Fold

DPSC FORM 2418
MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.